# The Adjacency-Local Uniqueness Lemma: Maxwell, Fierz–Pauli, and Yang–Mills on the Shell

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**Abstract**

Each gauge sector of the corpus makes a *uniqueness of the dynamics* claim: the relevant gauge-invariant action is forced, not chosen. For electromagnetism this is proved on the shell (the field paper's Theorem on the unique relevant Maxwell operator). For gravity the discrete Fierz–Pauli functional is exhibited and shown *exactly* gauge-invariant, but its uniqueness "within the adjacency-local class is expected from the continuum theorem and remains to be proven on the shell." This note proves it, and in doing so isolates the single lemma the three sectors share. We state the *adjacency-local uniqueness lemma*: for a connection-type field with a linear gauge orbit, the space of translation-invariant, adjacency-local, hypercubic-invariant, gauge-invariant quadratic functionals is, at the relevant (leading-symbol) order, *one-dimensional* — the curvature-square Casimir — with all higher-derivative invariants irrelevant. The proof is by symbol transversality and is the same in every sector. We verify it float-free for the symmetric rank-two field: the four independent two-derivative quadratic invariants reduce under linearised-diffeomorphism invariance to a one-dimensional space spanned by $(-1, 2, -2, 1)$, the linearised Einstein–Hilbert (Fierz–Pauli) functional; the result is unchanged under the central-difference shell substitution $k_\mu \mapsto \sin k_\mu$, and fails for one-sided differences exactly because their lattice adjoint is not their negative. The Fierz–Pauli mass term $h_{\mu\nu}h^{\mu\nu} - h^2$ is recovered as the unique ghost-free tuning. The Maxwell instance reproduces the field paper's theorem and the Yang–Mills instance reduces to it per colour plus the uniqueness of the Killing form. This discharges the "unique dynamics" clause of all three sectors at once. Verified in `validation/fierz_pauli_uniqueness.py` (10/10).

## 1 The residue

The gravitational field equation of this paper is built, not postulated: the field $h_{\mu\nu}$ is the frame perturbation, its gauge orbit $h_{\mu\nu} \mapsto h_{\mu\nu} + \nabla_\mu \xi_\nu + \nabla_\nu \xi_\nu$ is the linearised diffeomorphism gauge read off the relational ontology (Prop. *reframings are the gauge*), and the stiffness functional is the discrete Fierz–Pauli action

$$S[h] = \tfrac{\kappa}{2} \sum_x \Big( -\Delta_\lambda h_{\mu\nu}\, \Delta^\lambda h^{\mu\nu} + 2\, \Delta_\mu h^{\mu\nu}\, \Delta^\lambda h_{\lambda\nu} - 2\, \Delta_\mu h^{\mu\nu}\, \Delta_\nu h + \Delta_\mu h\, \Delta^\mu h \Big), \tag{1}$$

with $\Delta_\mu$ the central difference and $h = \eta^{\mu\nu} h_{\mu\nu}$. That (1) is *exactly* gauge-invariant on the periodic chart is verified (the central difference is anti-self-adjoint, $\Delta_\mu^\mathsf{T} = -\Delta_\mu$, with real symbol $\sin k_\mu$; one-sided differences are not, and break the invariance). What was *open* is uniqueness: that (1) is the *only* adjacency-local gauge-invariant stiffness, not merely *a* gauge-invariant one. The same clause is the open part of every sector's "unique dynamics" claim — proved for the vector (Maxwell) in the field paper, asserted for the tensor here, implicit for Yang–Mills. We close it generically.

## 2 The lemma

**Definition 1** (Adjacency-local quadratic functional)**.** On the periodic chart $\Lambda \simeq \mathbb{Z}_L^4$ let $\phi$ be a field carrying a linear representation of the point group, with a linear gauge orbit $\phi \mapsto \phi + G\xi$ for a first-order difference operator $G$ (the symmetrised gradient). A *quadratic functional* is $S[\phi] = \frac{1}{2}\sum_x \phi^{\mathsf{T}}\mathcal{K}\phi$ with $\mathcal{K}$ a translation-invariant, hypercubic-invariant kernel of bounded range; it is *range one* (adjacency-local) if $\mathcal{K}$ couples only nearest cells. Its *symbol* $\widehat{\mathcal{K}}(k)$ is a trigonometric polynomial; the *relevant order* is its leading homogeneous part as $k \to 0$.

**Lemma 1** (Adjacency-local uniqueness)**.** *Let $\phi$ be a connection-type field (a vector, a symmetric two-tensor, or a Lie-algebra-valued vector) with the corresponding linear gauge orbit. At the relevant order the space of gauge-invariant adjacency-local quadratic functionals is* one-dimensional*, spanned by the quadratic Casimir of the curvature:*

$$\textit{spin } 1: \ F_{\mu\nu}F^{\mu\nu}; \qquad \textit{spin } 2: \ \textit{linearised } R \ (\textit{Fierz–Pauli}); \qquad \textit{Yang–Mills}: \ \mathrm{Tr}\, F_{\mu\nu}F^{\mu\nu}.$$

*Higher-derivative gauge invariants exist but are irrelevant, suppressed by powers of the resolution floor. Gauge invariance restores full rotational invariance at the relevant order, annihilating every hypercubic anisotropy.*

*Mechanism.* Gauge invariance of $S$ is transversality of the symbol with respect to the gauge orbit: $\widehat{\mathcal{K}}(k)\,\widehat{G}(k) = 0$. Because $\widehat{G}(k)$ is the symbol of a first-order difference (a single power of the momentum), at the relevant order transversality is a homogeneous linear condition on the finitely many degree-two hypercubic-invariant tensors available to $\widehat{\mathcal{K}}$. Solving it leaves a one-dimensional space, and the surviving symbol is the transverse projector onto the curvature — the unique combination annihilated by a longitudinal (pure-gauge) momentum. The three sectors differ only in the index structure of $\phi$ and $\widehat{G}$; the linear algebra is identical. The sector-by-sector solutions follow. □

## 3 The three instances

**Spin one (Maxwell).** For a vector $A_\mu$ with $\delta A_\mu = \Delta_\mu\lambda$ the two-derivative quadratic invariants are $k^2(A\cdot A)$ and $(k\cdot A)^2$; transversality forces the combination $k^2(A\cdot A) - (k\cdot A)^2 = \frac{1}{2}F_{\mu\nu}F^{\mu\nu}$, unique. This is the field paper's theorem on the unique relevant Maxwell operator, recovered here as the spin-one instance of Lemma 1. (At range one a single irrelevant $k^4$ companion also survives, suppressed by two powers of the floor.)

**Spin two (Fierz–Pauli).** For the symmetric $h_{\mu\nu}$ with $\delta h_{\mu\nu} = \Delta_\mu\xi_\nu + \Delta_\nu\xi_\mu$ the complete basis of two-derivative quadratic invariants, modulo total differences, is four-dimensional:

$$L_1 = \Delta_\lambda h_{\mu\nu}\Delta^\lambda h^{\mu\nu}, \quad L_2 = \Delta_\mu h^{\mu\nu}\Delta^\lambda h_{\lambda\nu}, \quad L_3 = \Delta_\mu h^{\mu\nu}\Delta_\nu h, \quad L_4 = \Delta_\mu h\,\Delta^\mu h.$$

**Proposition 1** (Spin-two uniqueness)**.** *Transversality $\widehat{\mathcal{K}}(k)$ $\mathrm{sym}(k\,\xi) = 0$ for all $\xi$ leaves a one-dimensional space of coefficients $(a_1, a_2, a_3, a_4)$, spanned by $(-1, 2, -2, 1)$: the discrete linearised Einstein–Hilbert functional* (1)*. The condition is fourteen homogeneous equations of rank three on four unknowns; the nullity is one (verified by exact rational linear algebra).*

The solution is the unique gauge-invariant two-derivative quadratic stiffness: (1) is forced, not chosen, which is the clause that was open. The *massive* completion adds the no-derivative term $b_1 h_{\mu\nu}h^{\mu\nu} + b_2 h^2$; the coefficient of $h_{00}^2$ is exactly $b_1 + b_2$, so it vanishes — leaving $h_{00}$ a Lagrange multiplier that enforces the Hamiltonian constraint and removes the scalar ghost — iff $b_1 + b_2 = 0$, i.e. the mass term is the Fierz–Pauli combination $h_{\mu\nu}h^{\mu\nu} - h^2$. Uniqueness of the kinetic term and of the ghost-free mass term thus have the same source: gauge invariance, exact, and the residual constraint it leaves.

**Yang–Mills.** For a Lie-algebra-valued connection $A^a_\mu$ the linearised gauge orbit is $\delta A^a_\mu = \Delta_\mu \lambda^a$ — the Maxwell orbit, per colour. The spin-one instance therefore applies index by index and fixes the kinetic functional to $\sum_a F^a_{\mu\nu} F^{a\,\mu\nu}$ up to the choice of invariant bilinear on the algebra; on a simple Lie algebra that bilinear is the Killing form, unique up to scale, so the functional is $\mathrm{Tr}\, F_{\mu\nu} F^{\mu\nu}$, unique. The cubic and quartic self-couplings are then not free either: they are forced by promoting the linear orbit to the full non-abelian gauge transformation (the curvature must close), the content of the field paper's gluon self-coupling. On the lattice this functional is the leading term of the Wilson plaquette action, whose higher-representation completions are the irrelevant companions of Lemma 1.

## 4 On the shell

The proof is intrinsically a shell statement. The discrete functional's symbol is obtained from the continuum one by $k_\mu \mapsto \sin k_\mu$ in each central difference, and the gauge-orbit symbol carries the same substitution. The transversality conditions are polynomial identities in the symbol components that must hold for all lattice momenta; since the reachable values $(\sin k_0, \ldots, \sin k_3)$ fill a region with nonempty interior, the identities force the same coefficient solution as in the continuum. Hence Proposition 1 holds verbatim on the shell, with no $\Omega \to \infty$ limit: the nullspace is one-dimensional under the substitution $k_\mu \mapsto \sin k_\mu$, and equals $(-1, 2, -2, 1)$.

The role of the *central* difference is not cosmetic. The integration by parts that proves gauge invariance needs $\Delta_\mu^{\mathsf{T}} = -\Delta_\mu$; on the cyclic chart the central difference (symbol $i \sin k_\mu$, purely imaginary) is anti-self-adjoint, while the one-sided difference (symbol $e^{ik_\mu} - 1$) is not, and the invariance — hence the uniqueness argument built on it — fails for it. Both facts are verified directly.

## 5 What this closes

- **Gravity.** Prop. *coupling, conservation, and the imported uniqueness*: the clause "uniqueness within the adjacency-local class … remains to be proven on the shell" is now proven (Prop. 1). The functional (1) is the unique relevant gauge-invariant stiffness; the Status section's corresponding open item is discharged.
- **Electromagnetism.** The field paper's unique-Maxwell theorem is the spin-one instance of Lemma 1; the two now stand on one footing.
- **Yang–Mills.** The colour kinetic term $\mathrm{Tr}\, F^2$ is fixed by the same lemma (per-colour transversality + Killing form), the self-couplings by closure of the orbit.

One lemma, proved once by symbol transversality, discharges the "unique dynamics" clause of all three gauge sectors. What it does *not* fix is the overall normalisation of each functional — the coupling constant — which is the separate order-one/$\Omega$-hard question (gravity's $\kappa$ and $c'_S$, electromagnetism's $\alpha_{\mathrm{bare}}$, the strong scale): the lemma fixes the *form* of the dynamics uniquely, leaving exactly one number per sector.

**Reproduction.** `validation/fierz_pauli_uniqueness.py` (10/10): the spin-two nullspace is one-dimensional and equals $(-1, 2, -2, 1)$ in the continuum and under $k_\mu \mapsto \sin k_\mu$; central differences are anti-self-adjoint and one-sided are not; the Fierz–Pauli mass tuning is $b_1 + b_2 = 0$; and the spin-one cross-check returns the unique $F^2$.

## References


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